

Intercept Form

Vertex

Axis of Symmetry

Standard Form: $y = ax^2 + bx + c$

$\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

$x = \frac{-b}{2a}$

Vertex Form: $y = a(x - h)^2 + k$

(h, k)

$x = h$

Intercept Form: $y = a(x - p)(x - q)$

$\left(\frac{p+q}{2}, f\left(\frac{p+q}{2}\right)\right)$

$x = \frac{p+q}{2}$

****x-intercepts are $(p, 0)$ and $(q, 0)$**

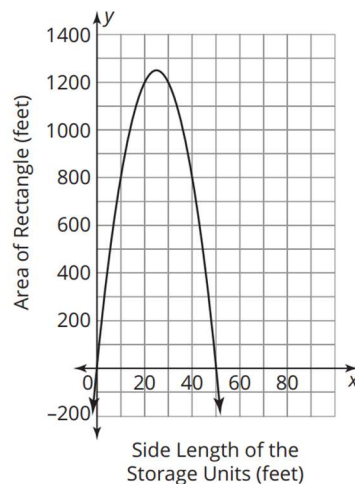
Some textbooks use the term Factored Form, this is similar to Intercept Form.

Example 1: Given the graph of a quadratic function and two equations of that quadratic function. What are the similarities and differences of the two given functions?

Area of Dog Enclosure

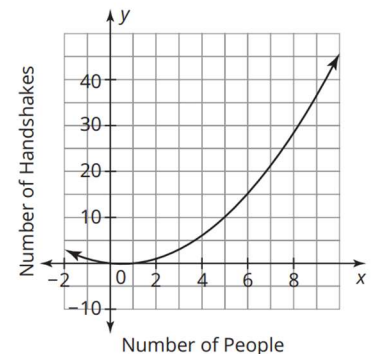
$$A(s) = -2s^2 + 100s$$

$$= -2(s)(s - 50)$$

**Handshake Problem**

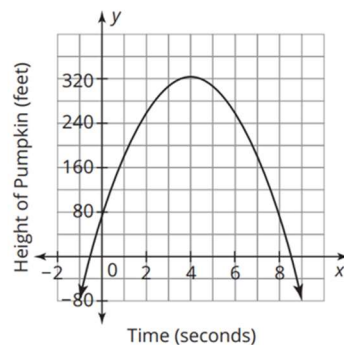
$$f(n) = \frac{1}{2}n^2 - \frac{1}{2}n$$

$$= \frac{1}{2}(n)(n - 1)$$

**Punkin' Chunkin'**

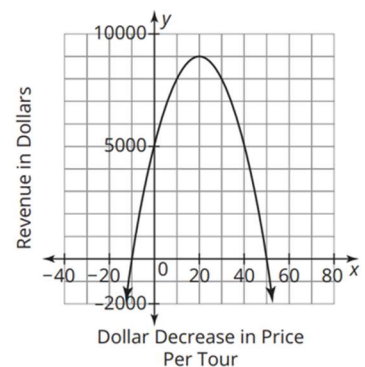
$$h(t) = -16t^2 + 128t + 68$$

$$= -16\left(t - \frac{17}{2}\right)\left(t + \frac{1}{2}\right)$$

**Ghost Tour**

$$r(x) = -10(x + 10)(x - 50)$$

$$= -10x^2 + 400x + 5000$$



Example 2: Determine the x-intercepts of each function. Find the vertex and axis of symmetry. Determine whether the vertex is a maximum or minimum. State the Domain and Range.

A) $h(x) = -2(x+3)(x+5)$

B) $b(x) = \frac{1}{2}(x+50)(x-150)$

C) $f(x) = \frac{1}{3}(x-1)(x-1)$

D) $j(x) = -2x(1-x)$

Example 3: Graph $h(x) = -2(x+3)(x+5)$

Example 4: Graph $j(x) = -2x(1-x)$

